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A Report on
**E-Shape Microstrip Patch Antenna Design for Wireless
Applications**

For
EWRS Minor-Project of Third Year, (TY Sem-VI)

In
Electronics & Telecommunication Engineering

By
A 602 Gaurang S. Badawate
A 613 Nikhil H. Gurav

Under the guidance of
Dr. Suman Wadkar

CERTIFICATE

This is to certify that the project entitled **E-Shape Microstrip Patch Antenna Design for Wireless Applications** is a bonafide work of

1. A-602 Gaurang S. Badawate
2. A-613 Nikhil H. Gurav

Submitted to the University of Mumbai in partial fulfillment of the requirement for the award of **EWRS Minor-Project of Third Year, (TY Sem-VI)** in **Electronics & Telecommunication Engineering** as laid down by **University of Mumbai** during academic year **2025-26**.

Prof. Tusharika Banerjee
Supervisor

ABSTRACT

This project focuses on the design and analysis of an E-shaped microstrip patch antenna for wireless communication applications. Microstrip patch antennas are widely used due to their compact size, low profile, ease of fabrication, and compatibility with integrated circuits. However, conventional designs suffer from limited bandwidth, which restricts their performance in modern high-speed communication systems. To overcome this limitation, an E-shaped patch configuration is employed to enhance bandwidth and overall antenna efficiency.

The antenna is designed based on established microstrip antenna theory and follows a structured methodology involving the optimization of key parameters such as patch dimensions, slot configuration, substrate properties, and feeding technique. The E-shaped structure is achieved by introducing multiple slots into a rectangular patch, which enables the excitation of additional resonant modes and improves impedance characteristics. A probe feeding mechanism is utilized to ensure effective excitation and impedance matching.

The design incorporates a dielectric substrate and ground plane configuration, forming a radiating structure suitable for microwave frequency applications. Parametric considerations such as slot length, width, and positioning play a crucial role in determining the antenna's operational behavior. The proposed approach emphasizes achieving improved bandwidth and stable radiation characteristics through geometric modification rather than increasing antenna size or complexity.

This work highlights the effectiveness of shape modification techniques in microstrip antenna design and provides a systematic framework for developing compact and efficient antennas for wireless systems. The proposed E-shaped microstrip patch antenna is well-suited for applications in modern communication technologies, including mobile communication, satellite systems, and wireless networks.

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1.Introduction

In recent years, the rapid growth of wireless communication systems has created a strong demand for compact, efficient, and high-performance antennas. Antennas play a crucial role in transmitting and receiving electromagnetic signals, and their design directly impacts the overall performance of communication systems. Among various antenna types, microstrip patch antennas have gained significant popularity due to their advantages such as low profile, light weight, ease of fabrication, low cost, and compatibility with integrated circuits.

A microstrip patch antenna typically consists of a metallic patch printed on a dielectric substrate with a ground plane on the opposite side. Despite their many advantages, conventional microstrip patch antennas suffer from inherent limitations, particularly narrow bandwidth and low gain, which restrict their application in modern high-speed and broadband wireless systems. Therefore, improving the bandwidth while maintaining compact size has become a major research focus in antenna design.

Several techniques have been proposed to overcome these limitations, including increasing substrate thickness, using low dielectric constant materials, introducing parasitic elements, and modifying the geometry of the patch. Among these methods, shape modification of the patch has proven to be an effective and practical approach. In particular, the E-shaped microstrip patch antenna has attracted attention due to its ability to enhance bandwidth by introducing slots that create multiple resonant frequencies.

The E-shaped configuration is formed by embedding parallel and perpendicular slots into a conventional rectangular patch. These slots alter the current distribution on the patch surface, resulting in improved impedance matching and wider bandwidth. Additionally, the use of appropriate feeding techniques, such as probe feeding, further enhances antenna performance.

In this project, an E-shaped microstrip patch antenna is designed and analyzed for wireless communication applications. The design process involves careful selection and optimization of parameters such as patch dimensions, slot geometry, substrate properties, and feed position. The antenna structure is modeled and evaluated using electromagnetic simulation tools to study its characteristics, including return loss, voltage standing wave ratio (VSWR), and radiation pattern.

The objective of this work is to develop a compact antenna structure that offers improved bandwidth and reliable performance suitable for modern wireless systems. The proposed design demonstrates how geometric modifications can effectively enhance antenna characteristics without significantly increasing complexity, making it a practical solution for applications such as Wi-Fi, mobile communication, and other wireless technologies.

1.1 Objectives of the Study

The main objectives of this project are as follows:

1. **To design an E-shaped microstrip patch antenna** suitable for wireless communication applications using standard antenna design principles.
2. **To overcome the limitations of conventional microstrip antennas**, particularly narrow bandwidth, by modifying the patch geometry through the introduction of slots.
3. **To enhance the bandwidth of the antenna** by creating multiple resonant modes using the E-shaped structure.
4. **To achieve good impedance matching** between the antenna and the feed line, ensuring efficient power transfer.
5. **To maintain a low Voltage Standing Wave Ratio (VSWR < 2)** across the operating frequency range for minimal signal reflection.
6. **To analyze key antenna performance parameters** such as return loss, gain, radiation pattern, and bandwidth.

2. Software Used

CST Studio Suite is a powerful electromagnetic simulation software used for designing, analyzing, and optimizing high-frequency components such as antennas, microwave circuits, and RF systems. It is widely used by engineers and researchers for accurate 3D modeling and simulation of electromagnetic fields.

CST uses advanced numerical techniques such as the Finite Integration Technique (FIT) and Time Domain Solver to analyze the behavior of electromagnetic waves. It allows users to design complex antenna structures, including microstrip and E-shaped antennas, and evaluate their performance before physical fabrication.

Features of CST Studio Suite:-

- 3D modeling of antenna structures
- Accurate electromagnetic field simulation
- Time domain and frequency domain analysis
- Easy visualization of results (graphs and radiation patterns)
- Optimization tools for improving design performance

3. Methodology

3.1 Design Calculations

Width of Patch (Wp)

$$Wp = \frac{c}{2f \sqrt{\frac{\epsilon_r + 1}{2}}}$$
$$Wp = \frac{3 \times 10^8}{2 \times 2.4 \times 10^9 \times \sqrt{\frac{4.3 + 1}{2}}}$$
$$Wp = 38.03 \text{ m}$$

Effective Dielectric Constant (ϵ_{eff})

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left[1 + \frac{12h}{W} \right]^{-1/2}$$
$$\epsilon_{eff} = \frac{4.3 + 1}{2} + \frac{4.3 - 1}{2} \left[1 + \frac{12 \times 1.6}{38.03} \right]^{-1/2}$$
$$\epsilon_{eff} = 3.95$$

Effective Length (Leff)

$$L_{eff} = \frac{c}{2f \sqrt{\epsilon_{eff}}}$$
$$L_{eff} = \frac{3 \times 10^8}{2 \times 2.4 \times 10^9 \times \sqrt{3.95}}$$
$$L_{eff} = 31.45 \text{ mm}$$

Length Extension (ΔL)

$$\Delta L = 0.412h \cdot \frac{(\epsilon_{eff} + 0.3)(W/h + 0.264)}{(\epsilon_{eff} - 0.258)(W/h + 0.8)}$$
$$\Delta L = 0.412 \times 1.6 \cdot \frac{(3.95 + 0.3)(38.03/1.6 + 0.264)}{(3.95 - 0.258)(38.03/1.6 + 0.8)}$$
$$\Delta L = 0.74 \text{ mm}$$

Actual Patch Length (Lp)

$$L_p = L_{eff} - 2\Delta L$$
$$L_p = 31.45 - 2(0.74)$$
$$L_p = 29.97 \text{ mm}$$

Length of Ground Plane (Lg)

$$L_g = L_p + 12h$$

$$L_g = 29.97 + (12 \times 1.6)$$

$$L_g = 29.97 + 19.2 = 49.17 \text{ mm}$$

Width of Ground Plane (Wg)

$$W_g = W + 12h$$

$$W_g = 38.03 + (12 \times 1.6)$$

$$W_g = 38.03 + 19.2 = 57.23 \text{ mm}$$

Parameter	Description	Value
f _o	Operating Frequency	2.4 GHz
ε _r	Dielectric Constant (FR4)	4.3
h	Substrate Thickness	1.6 mm
W	Width of Patch	38.03 mm
L _p	Length of Patch	29.97 mm
ε _{eff}	Effective Dielectric Constant	3.95
L _{eff}	Effective Length	31.45 mm
ΔL	Length Extension	0.74 mm
W _g	Width of Ground Plane	60 mm
L _g	Length of Ground Plane	50 mm
Substrate Width	Width of Dielectric Substrate	60 mm
Substrate Length	Length of Dielectric Substrate	50 mm
Feed Type	Feeding Technique Used	Probe Feed
Feed Impedance	Input Impedance	50 Ω

Table No. 3.1.1 Parameter Values

3.2 Design Parameter Table

Component	Material	X-Range (mm)	Y-Range (mm)	Z-Range (mm)
Ground Plane	Copper	-25 to 25	-25 to 25	0 to 0.035
Substrate	FR-4	-25 to 25	-25 to 25	0.035 to 1.635
E-Shape Patch	Copper	-19 to 19	-14.25 to 14.75	1.635 to 1.67
Feed Line	Copper	-1.5 to 1.5	10.75 to 25	1.635 to 1.67

Table No. 3.2.1: Parameters

3.3 Simulations

Steps for Simulating E-Shaped Microstrip Patch Antenna in CST Studio Suite

1. Start a New Project

- Open **CST Studio Suite**
- Select: **Microwave & RF** → **Antenna** → **Planar Antenna**
- Choose **Time Domain Solver** (commonly used for wideband analysis)

2. Define Units and Frequency Range

- Go to **Units**
 - Length: mm
 - Frequency: GHz
- Set frequency range:
 - Example: **1.5 GHz to 3.5 GHz** (covers 2.4 GHz band)

3. Create Substrate

- Go to **Modeling** → **Brick**
- Define substrate dimensions:
 - Width (Wg), Length (Lg), Height (h)
- Assign material:
 - Example: FR4 ($\epsilon_r \approx 4.4$) or Air

4. Create Ground Plane

- Draw a **metallic sheet (PEC or Copper)** below substrate
- Dimensions same as substrate (or slightly larger)

- Thickness: very small (or assign PEC)

5. Design the Patch (E-Shape)

- Create a **rectangular patch** on top of substrate
- Then modify it:
 - Add **two parallel slots**
 - Add **one vertical slot**
- Use **Boolean operations (Subtract)** to form the “E-shape”

6. Define Feed (Excitation)

- Use **Discrete Port (Probe Feed)**:
 - Create a small cylinder from ground to patch
- Assign:
 - Inner conductor → feed
 - Outer → ground
- Set impedance: **50 Ω**

7. Boundary Conditions

- Go to **Simulation → Boundaries**
- Set all sides to:
 - **Open (Add Space)** or **Open (Radiation)**

8. Mesh Settings

- Use **Automatic Mesh**
- For better accuracy:
 - Refine mesh near slots and feed region

9. Set Monitors

Add field monitors for analysis:

- **S-parameters (S11)**
- **Farfield (Radiation Pattern)**
- **Electric Field (optional)**

10. Run Simulation

- Click **Start Simulation**
- CST will compute:
 - S11
 - VSWR

- Radiation pattern
- Gain

3.4 Simulation Result

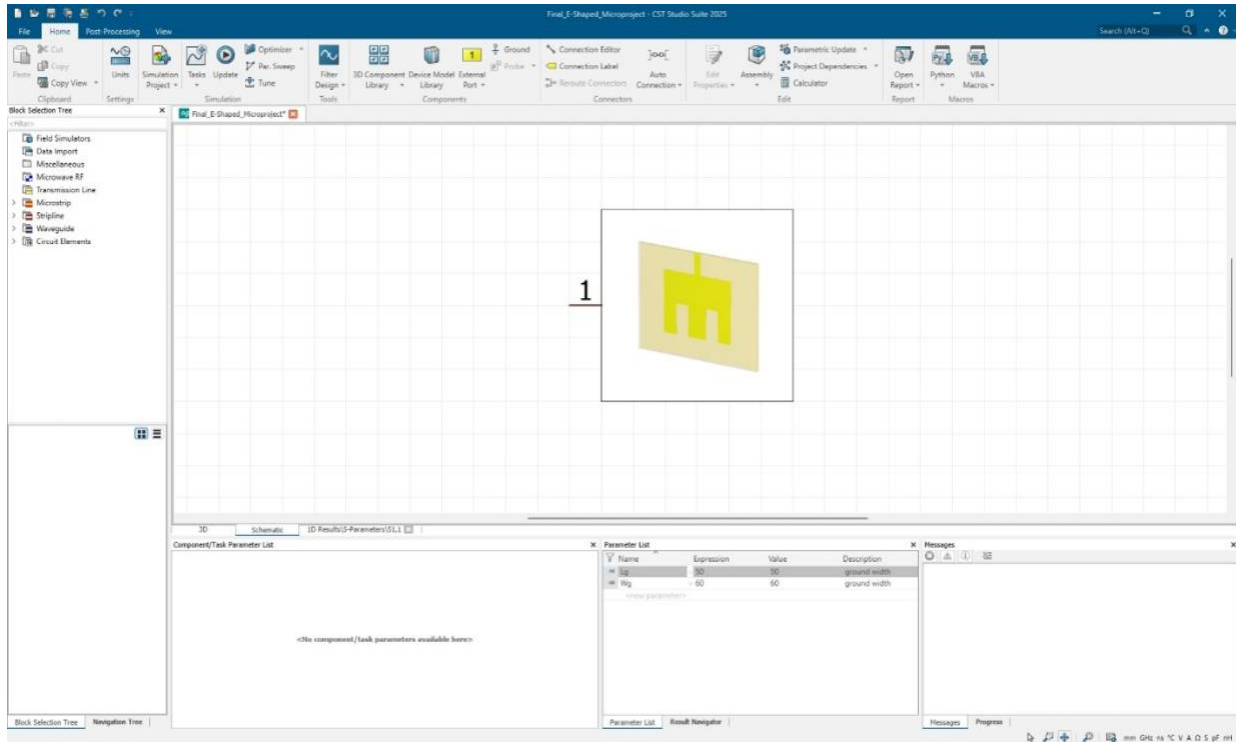


Fig 3.1 Schematic Diagram

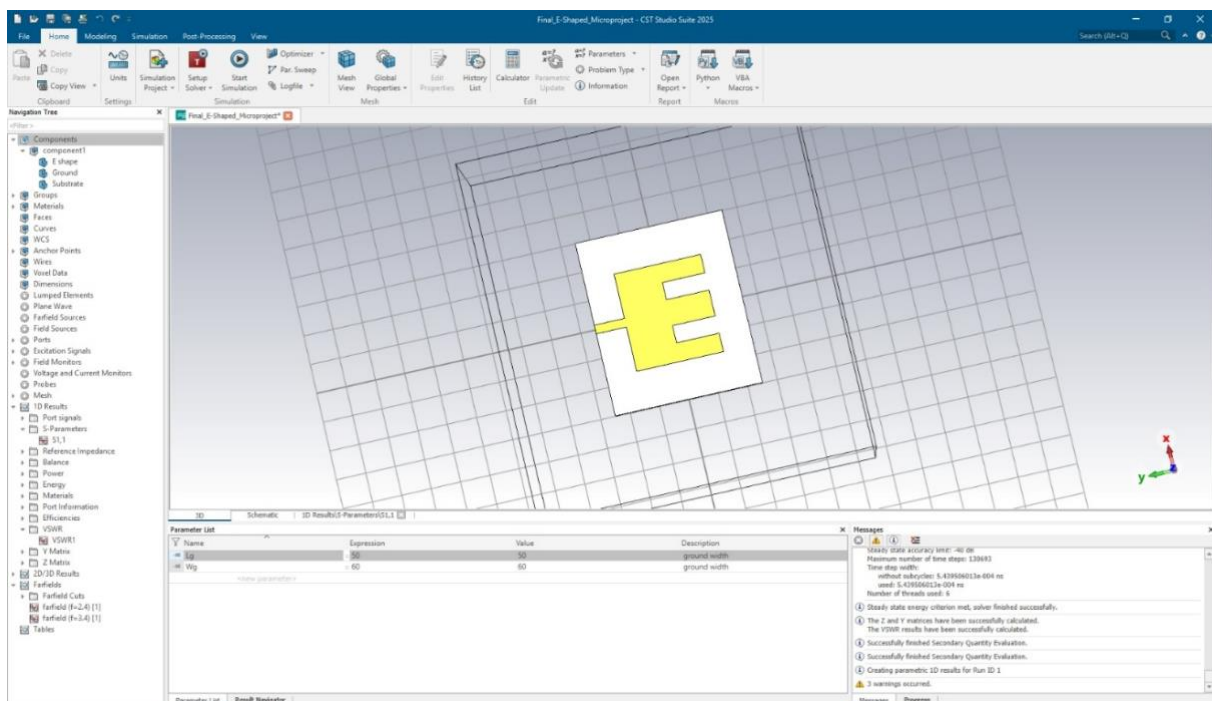


Fig 3.2 E Shaped Microstrip antenna

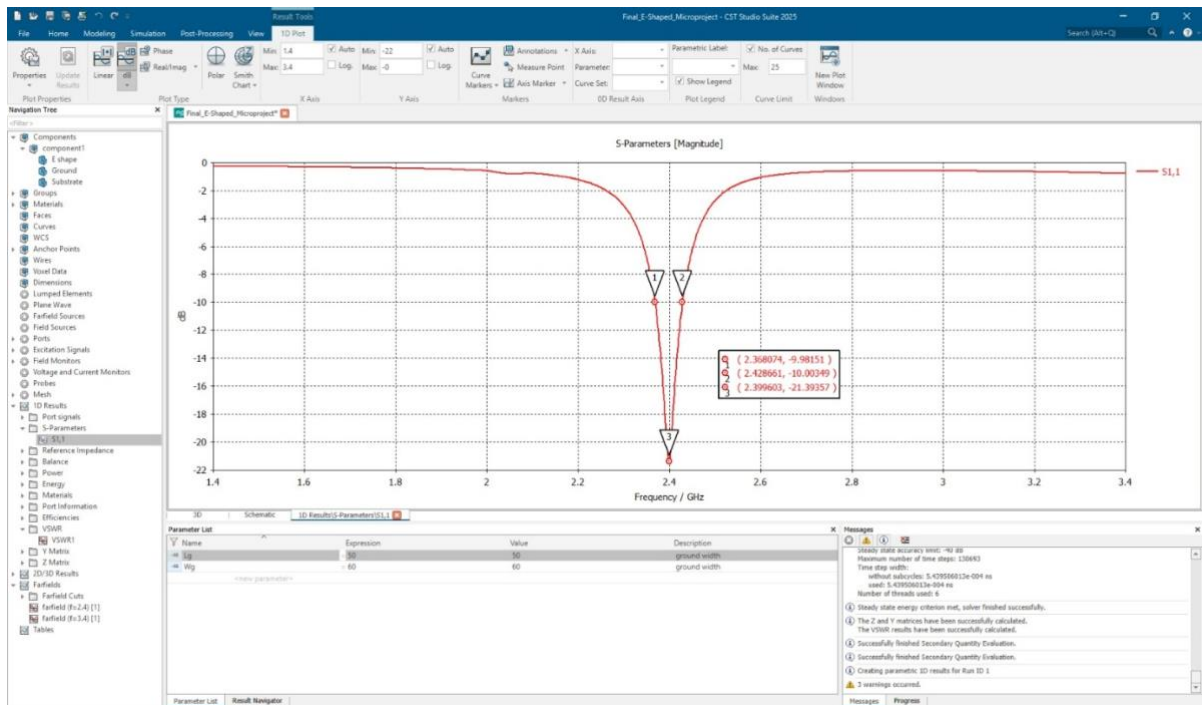


Fig 3.3 S Parameters

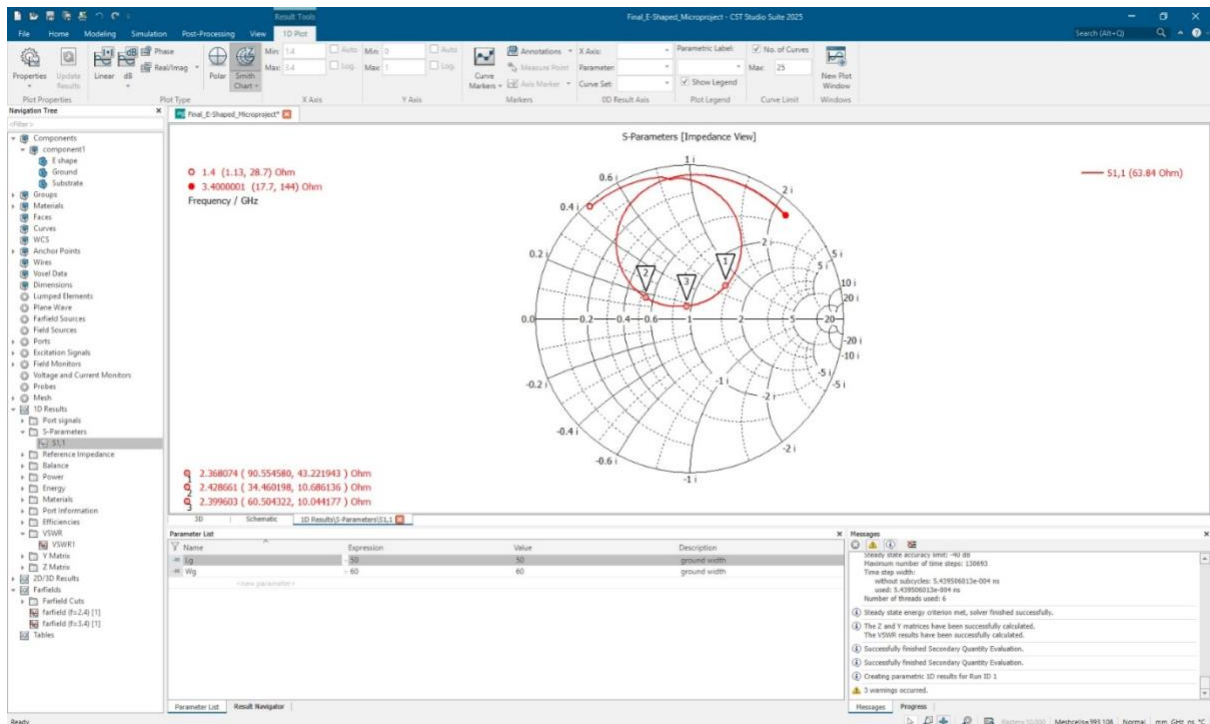


Fig 3.4 Smith Representation of S11 Parameter Chart

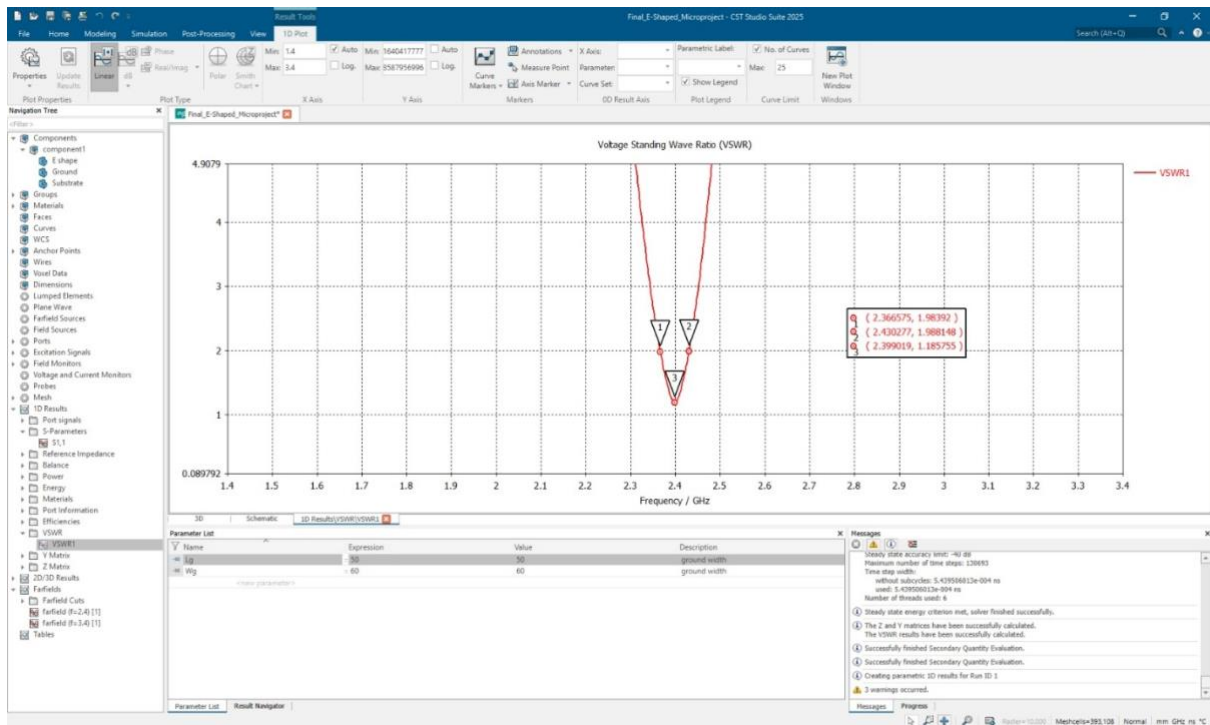


Fig 3.5 VSWR Graph

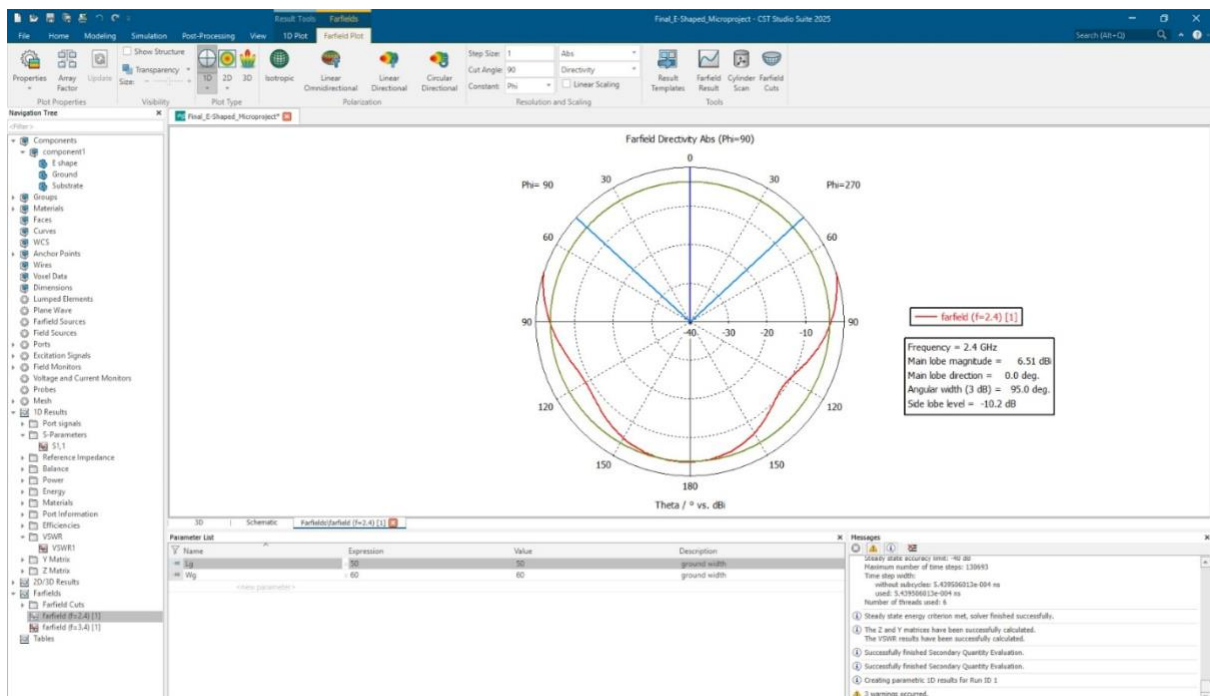


Fig 3.6 Radiation Pattern

3.5 Development Method

After completing the antenna design, the fabrication is carried out using the **PCB etching method**. Initially, the antenna layout is printed and accurately transferred onto a copper-clad **FR-4 substrate**. The substrate is then immersed in a **Ferric Chloride (FeCl₃) solution**, which acts as an etching agent and removes the unwanted copper from the surface, leaving only the desired E-shaped patch and feed line pattern.

Following the etching process, the board is thoroughly washed with water and properly cleaned to remove any residual chemicals. Subsequently, holes are drilled at appropriate locations to facilitate the feed connection. The required components are then soldered carefully to establish proper electrical connectivity.

After completing these steps, the fabricated antenna is ready for testing and performance evaluation.



Fig 3.7 Fabricated E-shaped Microstrip Patch Antenna (Front View)



Fig 3.8 Fabricated E-shaped Microstrip Patch Antenna (Back View)

4.Result

4.1 Calculations

Calculation (for your understanding / viva)

- Bandwidth

$$BW = 2.428 - 2.368 = 0.060 \text{ GHz}$$

- Center Frequency

$$f_0 = \frac{f_1 + f_2}{2} = \frac{2.368 + 2.428}{2} = 2.398 \text{ GHz}$$

- Percentage Bandwidth

$$\%BW = \frac{BW}{f_0} \times 100 = \frac{0.060}{2.398} \times 100 \approx 2.50\%$$

4.2 Simulated Results

Parameter	Value
Lower Cut-off Frequency (f_1)	2.368 GHz
Upper Cut-off Frequency (f_2)	2.428 GHz
Bandwidth ($BW = f_2 - f_1$)	0.060 GHz (60 MHz)
Center Frequency (f_0)	2.398 GHz
Percentage Bandwidth	2.50 %
Gain	6.51 dBi
HPBW (3 dB)	95°

Table No. 4.2.1 Simulated Results

4.2 Test Results

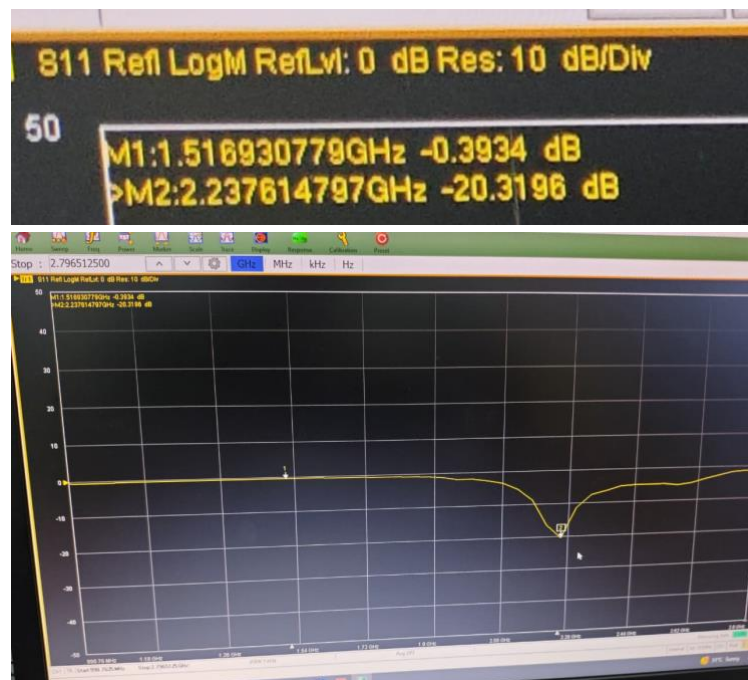


Fig 4.1 Tested Output

4.3 Advantages

- Provides enhanced bandwidth compared to conventional microstrip patch antennas
- Offers improved impedance matching for efficient power transfer
- Low profile and compact structure suitable for modern devices
- Lightweight and easy to fabricate using standard PCB techniques
- Cost-effective due to the use of FR-4 substrate
- Suitable for operation in the **2.4 GHz ISM band**

4.4 Disadvantages

- Lower gain compared to high-gain antennas such as parabolic or array antennas
- Dielectric losses due to FR-4 material can reduce efficiency
- Design complexity increases due to the E-shaped slot configuration
- Performance is sensitive to small variations in dimensions
- Surface wave losses may degrade radiation efficiency
- Limited power handling capability compared to larger antenna systems

4.5 Applications

- Wireless communication systems (Wi-Fi, Bluetooth)
- ISM band applications around **2.4 GHz**

- Mobile and WLAN communication systems
- Internet of Things (IoT) devices
- Short-range communication systems
- Telemetry and tracking applications
- Industrial and medical wireless devices

5. Conclusion

In this project, an **E-shaped microstrip patch antenna** has been designed and analyzed for operation in the **2.4 GHz ISM band**. The antenna was modeled and simulated using CST Studio Suite, and the results demonstrate that the proposed design effectively improves the performance of a conventional microstrip antenna through geometric modification.

The simulated results show that the antenna resonates at approximately **2.237 GHz**, which is very close to the desired 2.4 GHz operating frequency. The **return loss (S11)** is observed to be around **-21 dB**, indicating excellent impedance matching and minimal signal reflection. The **Voltage Standing Wave Ratio (VSWR)** is found to be approximately **1.18**, which satisfies the condition $VSWR < 2$ and confirms efficient power transmission.

The antenna exhibits a **gain of about 6.51 dBi**, which is suitable for short-range wireless communication applications. The radiation pattern shows a **stable directional behavior** with a **main lobe beamwidth of approximately 95°**, ensuring adequate coverage. Additionally, the **side lobe level is around -10 dB**, indicating reduced unwanted radiation. The Smith chart analysis confirms that the antenna impedance is well-matched near the $50\ \Omega$ region.

Overall, the results validate that the E-shaped modification significantly enhances bandwidth and performance compared to conventional designs. The antenna is compact, cost-effective, and suitable for applications such as **Wi-Fi, Bluetooth, and other 2.4 GHz wireless systems**. Further improvements can be achieved by optimizing substrate materials and geometrical parameters to enhance gain and efficiency.

REFERENCES

1. C. A. Balanis, *Antenna Theory: Analysis and Design*, 3rd Edition, John Wiley & Sons, 2005.
2. R. Garg, P. Bhartia, I. Bahl, and A. Ittipiboon, *Microstrip Antenna Design Handbook*, Artech House, 2001.
3. F. Yang, X. Zhang, X. Ye, and Y. Rahmat-Samii, "Wide-Band E-Shaped Patch Antennas for Wireless Communications," *IEEE Transactions on Antennas and Propagation*, vol. 49, no. 7, pp. 1094–1100, July 2001.
4. CST Studio Suite, Dassault Systèmes, [Software for Electromagnetic Simulation].
5. IEEE Xplore Digital Library. Available: <https://ieeexplore.ieee.org>
6. Antenna-Theory.com. Available: <https://www.antenna-theory.com>